For the teacher:

Activity 1: Give the students a bag of random household goods/lab equipment/ stationary etc and ask them to classify the goods into groups.

Get them to explain to the class how they did it.

You can get them to try as many ways as possible.

After second worksheet.

Activity 2: Give each student 6 different lollies (Minties, Fantales, Clinkers, Marshmallow, Jaffas and Bon Bons).

Place them on a piece of A3 paper and get them to start to make a dichotomous key to classify them.

Activity 3: Use the “Fishy Features” page to make a dichotomous key for the marine species listed.

Activity 4: Hand the students a copy of the ‘Fish Dichotomous Key’. Show them one fish at a time and ask them to identify them from the key.

Classification

When classifying organisms, taxonomists use the following framework:

Kingdom _____
Phylum _____
Class _____
Order _____
Family _____
Genus _____
Species _____

Classification makes it easier to identify, describe and remember species. Since there are thousands of organisms described from Australia alone, and more new species are being discovered every year, the use of an international classification system is extremely important.

1. What does classification mean? _____

2. Why do we classify organisms? _____

_____

Classification is used for the scientific naming, identifying and describing of individuals, and determining relationships between groups and their evolutionary links.

3. What does evolutionary links mean? _____

_____

Scientific names are internationally recognised names given to living things. Scientific names are always written in *italics*, with the genus name first, followed by the species name, e.g., *Plectropomus leopardus* is the scientific name for coral trout.

4. What is your scientific name? _____

Coral Trout



Classification



Taxonomists are scientists who classify and name plants and animals.

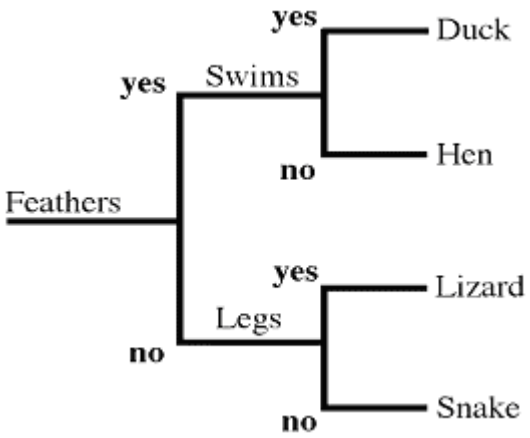
A Swedish botanist in 1753 by the name of Carolus Linnaeus devised a two-name system that we now know as binominal nomenclature. The first name is the organism's generic name or genus. This is also used in the name of closely related organisms. The second name is the organism's specific name or species.

1. What does the prefix 'bi' mean? (think bicycle, how many wheels?) _____

2. How many names in the binominal nomenclature system? _____

A dichotomous key is a method for identifying something by going through a series of choices leading to the correct answer. **Dichotomous means divided into 2 parts**. At each step, there are **two choices**, with each answer leading to another question with a further 2 choices until the item is identified.

Dichotomous Key



3. What does dichotomous mean?

_____

_____

When making a dichotomous key:

- Use **constant** characteristics rather than **variable** ones. Variable means _____
- Use **measurements** rather than terms like '**large**' and '**small**'.
- Use characteristics that are generally available to the user of the key rather than seasonal characteristics.
- Make the choice positive – something 'is' instead of 'is not'.
- Precede descriptive terms with the name of the part to which they apply.

Fishy Features

Common name	Distinguishing characteristics
1. Baldchin Groper 	White 'chin' • Greenish body with yellow/orange fins • Peg-like teeth
2. Swallowtail 	Reddish silver • Long swallowtail tail • Large eyes
3. Bight redfish 	Reddish silver • Large eye • Prominent white lateral line
4. Roe's abalone 	Grooved shell • Mollusc
5. King George whiting 	Long slender fish • Small brown spots on side • Grows much larger than other whiting species
6. Western School whiting 	Dark blotch at base of pectoral fin
7. Squid 	Eight arms and two tentacles • Mollusc
8. Western blue groper 	Blue to green in colour • Large fleshy lips • Peg-like teeth
9. Cobbler 	Eel-like tail • Barbels around mouth • Venomous spine in dorsal and pectoral fins
10. Banded sweep 	Black cross bands on body

Fish illustrations © R.Swainston/www.anima.net.au

Fish Dichotomous Key

Step 1	
If the fish has a lunate shape tail (like a crescent moon),	*go to Step 2*
If the fish doesn't have a lunate shape tail,	*go to Step 3*
Step 2	
If the fish has one dorsal fin shaped like a sail,	it is a sailfish
If the fish has two dorsal fins,	it is a blue-fin tuna
Step 3	
If the fish has eyes on top of its head,	it is a flounder
If the fish has eyes on the side of its head,	*go to Step 4*
Step 4	
If the fish has a head shaped like that of a horse	it is a seahorse
If the fish doesn't have a head shaped like that of a horse	*go to Step 5*
Step 5	
If the fish has mouth barbels (like whiskers),	it is an estuarine cobbler
If the fish doesn't have mouth barbels,	*go to Step 6*
Step 6	
If the fish has spines,	*go to Step 7*
If the fish doesn't have spines,	*go to Step 8*
Step 7	
If the fish has short spines all over body,	it is a globefish
If the fish has long spines on dorsal and pectoral fins,	it is a lionfish
Step 8	
If the fish has large potato-sized black blotches over body,	it is a potato cod
If the fish is reddish with round blue spots,	it is a coral trout



